Course AE4-135 Rotor and Wake Aerodynamics

Assignment II: Programming a Lifting Line Model

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TU Delft

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1
Introduction

1.1. Abstract

In this report, we build and benchmark a Lifting-line model for wind-rotor aerodynamics. This report includes the fundamental background on disk actuator theory and rotor disc theory, as well as the demonstrations of fundamental quantities for analyzing the aerodynamic performance of wind turbine rotors in the context of the Lifting-line theory. Here, we analyze the radial distribution of the fundamental performance parameters, such as induction factors, force, and power coefficients, for a 100-meter diameter wind-turbine rotor model at a fixed tip-speed ratio 8. The current analysis is carried out for varying wake geometries, time-stepping, and discretization sizings. We also compare the results obtained with implemented Lifting-line model with results from a Blade-element-momentum theory model.

1.2. Background

The simplest model for a rotor, e.g., a wind turbine or airplane propeller, is the disk actuator model, described initially by Betz [2]. In Betz's model, the rotor is idealized as an actuator disk, representing the area swept by an idealized rotor composed of infinite blades rotating at infinite angular velocity. By applying the conservation of mass, momentum, and energy to the flowing fluid mass upstream and downstream of the rotor disk, energy is extracted or added to the flow via a pressure drop or rise through the change of momentum of the flow due to the thrust at the actuator. The conservation of mass in the disk actuator model imposes that the streamlines that cross the disk do not interact with the surrounding flow and comprehend a region of the flow named stream tube. Moreover, since the only mechanism for energy interaction between the disk actuator and the flow is the thrust force, no energy loss is predicted in the disk actuator model. However, turbines operate under finite tip-speed ratios and angular speeds, yielding energy losses.

In a realistic rotor model, energy is extracted or added to the flow via torque; in this case, we say that the flow imparts a torque on the wind turbine, whereas thrust is simply a byproduct of torque. The torque the flow imparts on the rotor, according to Newton's third law, yields torque imparted from the rotor's blades to the fluid, causing the flow crossing the area swept by the rotor to rotate. Thus, the streamlines leaving the area swept by the rotor blades will present a rotational motion composed of axial and tangential components (i.e., parallel and perpendicular to the rotor's rotational axis, respectively). Blade-element-momentum (BEM) theory codes do not account for the extra induction caused by the rotation of the fluid wake, and corrections for wake rotations and a finite number of blades (also known as tip and root losses) are necessary.

The next step in terms of accuracy from the BEM model is the Lifting-line model (LLM), also known as Pradtl's Lifting-line model. The LLM is a mathematical model in aerodynamics that allows for predictions of lift distribution over three-dimensional winds based on the wings' and the wake's geometry. More specifically, in the LLM framework, the shape of the wake behind the immersed body is taken into account in the determination of the potential velocity field, thus providing more accurate predictions in situations where local aerodynamic characteristics strongly correlate with the dynamics [6].

This report described the most important mathematical and computational aspects of baseline LLM computations. The mathematical theory underlying the aerodynamics and geometrical characteristics

of an LLM code is presented in Section 1.2.1.

In Section 2.1, we present the geometric elements and the general philosophy of the implemented LLM code. The results obtained with the implemented LLM code are benchmarked with a 100-meter-diameter rotor in Section 3. In **??** and 3.2, some concluding remarks and recommendations regarding future developments are made.

1.2.1. Lifting-line thoery

A potential flow ($\nabla \cdot (\nabla \times \vec{u}) = 0 \mid \vec{u} = u_x\hat{i} + u_y\hat{j} + u_z\hat{k}$) flow field around a blade can be generally decomposed into a (far-field) potential flow and a (near-wall) boundary layer flow. At the boundary layer flow, viscous flow effects are dominant and produce large velocity-field gradients which, in their turn, yield high-vorticity velocity fields, i.e., $\vec{\omega} = \nabla \times \vec{u} \gg 0$. The far-field flow, on the other hand, is typically free of viscous effects and thus can be roughly approximated as an irrotational ($\vec{\omega} = \vec{0}$) flow. Therefore, in the context of potential-flow theory (see Chapter 3 of [1]), the flowfield around an arbitrary three-dimensional blade can be described in terms of the distribution of source flows, $\nabla \cdot \vec{u}$, and vorticity, $\vec{\omega}$, such as illustrated in the top panel of Fig. 1.1.

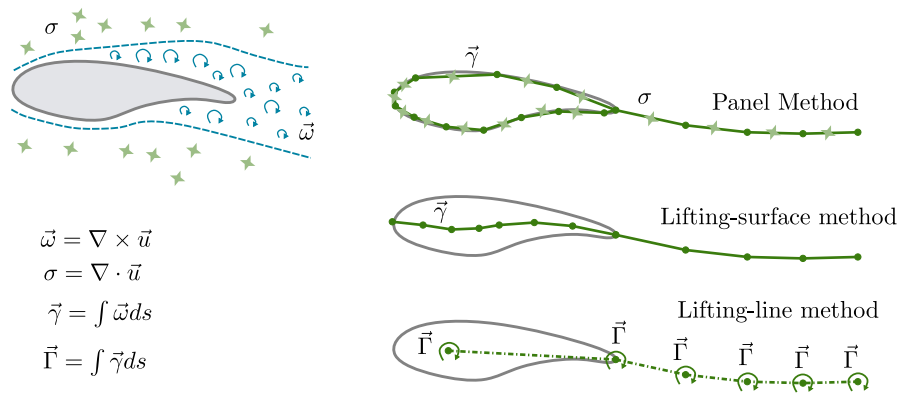


Figure 1.1: Flowfield approximation from three-dimensional to Lifting-line method.

As simplification measures, we can lump the vorticity around a body and in the wake into vortex filaments, as illustrated in the bottom panel of Fig. 1.2. With this simplification strategy, the original volume integrals necessary to solve the velocity field around a three-dimensional blade reduce to line integrals along the lumped vorticity filament along the body, as illustrated in the figure. More specifically, in the LLM framework, we lump the mean vorticity around the wing onto a single point at its quarter chord length.

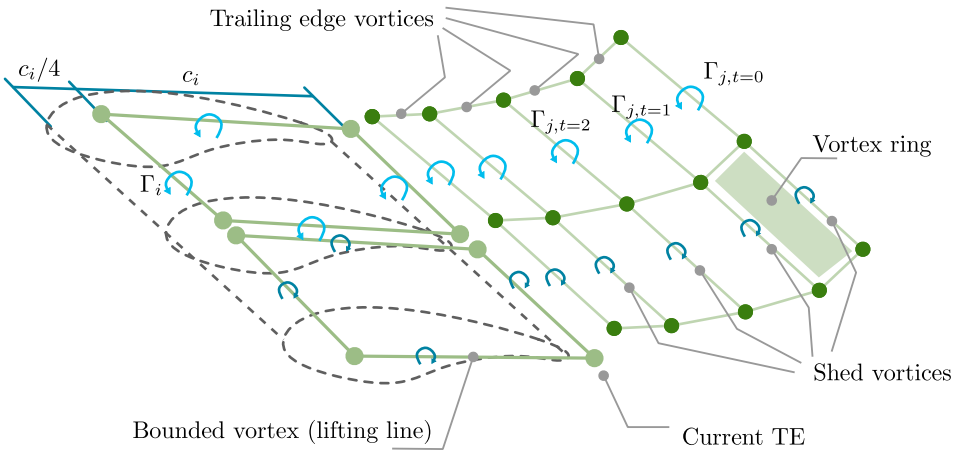


Figure 1.2: Wake geometry and lifting-line theory discretization philosophy.

From the Kelvin-Helmholtz's circulation theorem, we know that vortex tubes cannot have free ends in the fluid and must therefore be closed rings (see page 144 of [5]). In the context of LLM, Kelvin-Helmholtz's theorem imposes that the bounded vortex filament forms a loop with three other straight-line filaments downstream of the body, as illustrated in Fig. **??**. To respect Kelvin-Helmholtz's circulation theorem, the vortex strength, Γ, must also be constant along the loop. Hereafter, we refer to the four-interconnected vortex filaments as "vortex rings". Note also that the first filament of the vortex ring is, in fact, the bounded vortex filament.

Suppose we discretize each blade of a rotor into $j = 1, 2, \cdots$ panel sections. In that case, each panel section will be represented by their respective bounded vortex filament and will therefore shed a vortex-ring with the same vortex strength Γ_j, such as illustrated in Fig. **??**. In this case, each panel section will have a circulation strength Γ_j and will feel an incremental section lift force of δL_j (i.e., a lift-per-unit length) determined by the Kutta-Joukowski theorem:

$$\delta\vec{L}_j = \rho\vec{u} \times \vec{\Gamma}_j = \rho\Gamma_j(\vec{u} \times d\vec{l}_i). \tag{1.1}$$

In Eq. 1.1, ρ is the fluid density, and $d\vec{l}_i$ is the length of the i-th vortex filament. Notice that each filament $d\vec{l}_i$ induces an incremental contribution to the total lift force value $\vec{L}$, such that:

$$\vec{L}_j = \sum_i \delta\vec{L}_j(d\vec{l}_i) \tag{1.2}$$

Moreover, from the Biot-Savart law, we know that the velocity induced by a volume of vorticity lumped into a vortex filament $d\vec{l}_i$ on a point $\vec{x}_j$ of the flow is:

$$\vec{u}_\Gamma(\vec{x}_j) = \sum_i \left(-\frac{1}{4\pi} \int \Gamma_i \frac{\vec{r} \times d\vec{l}}{r^3} \right), \tag{1.3}$$

where $\vec{r}$ is the distance vector between the i-th filament and a j-th control point, $\vec{x}_j$, in the flow. For a straight-line segment filament with constant vortex strength Γ_i (see Fig. 1.3), the integral above can be expressed analytically as [6]:

$$\vec{u}_\Gamma(\vec{x}_j) = \sum_i \left(\frac{\Gamma_i}{4\pi} \frac{(r_1 + r_2)(\vec{r}_1 + \vec{r}_2)}{r_1 r_2 (r_1 r_2 + \vec{r}_1 \cdot \vec{r}_2)} \right). \tag{1.4}$$

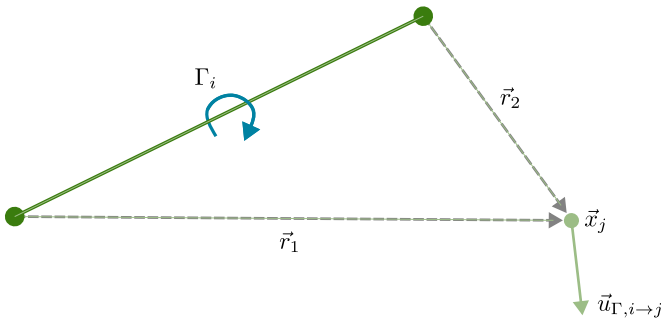


Figure 1.3: Vortex line geometry and induced velocity.

The total relative wind speed at a point $\vec{x}_j$ in the flow is simply the vectorial sum of the velocity $\vec{u}_{\Gamma,i}$ induced by the i-th vortex filament, the free-stream (source) velocity, $\vec{u}_\infty$, and the body-motion velocity, $\vec{u}_{rot}$, such as:

$$\vec{u}(\vec{x}_j) = \vec{u}_{rot}(\vec{x}_j) + \vec{u}_\infty + \sum_i \vec{u}_{\Gamma,i\to j}, \tag{1.5}$$

where $u_{\Gamma,i\to j}$ is the induced velocity of the i-th filament on the j-th control point . Notice that the wake behind the rotor's blade can consist of multiple vortex rings along the x-direction, each representing the overall vortex strength at a given time-step t of the flow. As we will further elaborate in Section **??**, the shape of a blade's wake, as well as the number of time steps (hence, the number of shed vortex rings) regarded in the Lifting-line model, are user-imposed; thus, we will also evaluate the impact of the selection of these variables on the Ligting-line model results. This methodology of imposing and fixing a wake design is called the "frozen-wake" methodology. The frozen wake methodology greatly simplifies the calculations in LLM by removing the costly calculations of self-induction velocities on the vorticity filaments representing the wake.

We highlight below the assumptions and limitations of the implemented LLM:

1. The flow is potential (incompressible and irrotational/inviscid)
2. The boundary layer vorticity is lumped into the quarter-chord length point; hence the method's accuracy level is above the blade-element-scale (thickness and displacement effects are therefore ignored),
3. Due to the adopted frozen-wake methodology, self-induction in the wake is not modeled
4. Three-dimensional flow effects are only considered in the wake, whereas force calculations on blade elements in derived from two-dimensional tabulated data. Thus, the method's accuracy relies upon two- and three-dimensional flow models.

Suppose the control points $\vec{x}_j$ are located at the aerodynamic centers of the blade sections. From Blade Element Theory, we can use the three components of $\vec{u}(\vec{x}_p)$ to determine the inflow angle, ϕ_j, angle-of-attack, α_j, and, thus, the aerodynamic lift force, $L(\alpha_j)$ acting on the j-th blade section (see Fig. **??**). We apply the Kutta-Joukowski formula to determine Γ_j from $L(\alpha_j)$, such that:

$$\Gamma_j = \frac{1}{2} W_j^2 C_{l,j}(\alpha_j) c_j, \tag{1.6}$$

where $W_j \equiv |\vec{W}_j \cdot \vec{W}_j| = |\vec{u}^k(\vec{x}_j) \cdot \vec{u}^k(\vec{x}_j)||$ is the magnitude of the perceived wind speed and c_j is blade section's chord length. Notice that we need Γ_j from Eq. 1.6 to determine the induced velocity $u_{\Gamma,i\to j}$. However, $u_{\Gamma,i\to j}$ is also needed to calculate the perceived wind speed, W_j in. Thus, the LLM consists ultimately of an iterative process to converge the bounded circulation value Γ_j until changes in $u_{\Gamma,i\to j}$ between consecutive iterations are sufficiently small.

1.2.2. Blade Element Theory

In the previous subsection, we presented the aerodynamics of the Lifting-line theory from the point of view of the flow, which is discretized into a free-stream source flow, $\vec{u}_\infty$, and multiple vortex filaments representing the bounded and shed vorticity in the wake. In this subsection, we will review the aerodynamic forces imparted by the Lifting-line theory onto the blade sections from the point-of-view of the blade sections. The primary simplification assumption in this subsection is that the flow at different sections of the rotor's blades is independent of the adjacent flows. Thus (tabulated) polar data from two-dimensional Blade Element Theory can be used to determine the aerodynamic forces imparted on the different blade sections.

From blade element theory, each airfoil section's lift, L_j (per-unit span) force vector, parallels the relative, local flow velocity vector, $\vec{W}_j$. From the two-dimensional Blade Element Theory, the magnitude of the lift force per unit span is given by:

$$L_j = \frac{1}{2} \rho W_j^2 c_j C_{l,j}(\alpha_j). \tag{1.7}$$

In the current LLM methodology, $C_{l,j}(\alpha_j)$ is determined by linear interpolating tabulated airfoil polar data. The magnitude of the axial, $F_{a,j}$, and tangential, $F_{t,j}$, forces acting on the the j-th blade section (see Fig. **??**) are computed using:

$$F_{a,j} = L_j \sin(\phi_j), \tag{1.8}$$
$$F_{t,j} = L_j \cos(\phi_j). \tag{1.9}$$

Notice that drag forces are ignored in the current LLM implementation. The total forces acting on the rotor's shaft are given by:

$$F_a = \sum_j F_{a,j}, \tag{1.10}$$
$$F_t = \sum_j F_{t,j}. \tag{1.11}$$

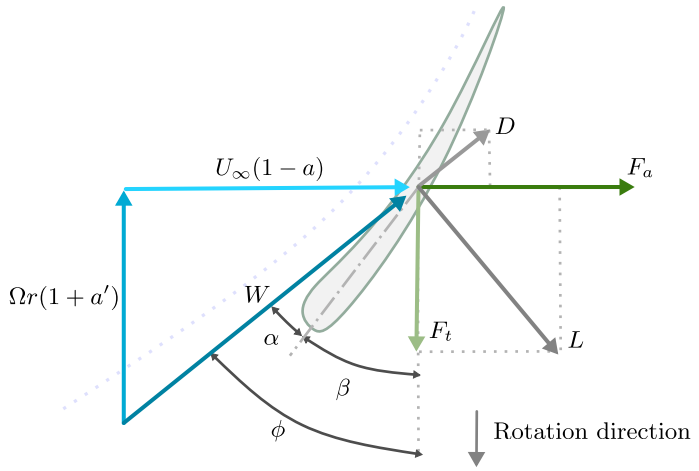


Figure 1.4: This schematic diagram shows the forces acting on each two-dimensional blade element and the angles between the velocity vectors and the axis of rotation. The aerodynamic forces are obtained from classical Blade Element Theory.

From Blade Element Momentum theory, the velocity at each blade section decreases by an induction factor of a_j relative to the free-stream speed flow, yielding a perceived axial velocity of $U_\infty \equiv |\vec{u}(-\infty) \cdot \vec{u}(-\infty)|$, $U_\infty(1-a)$. Moreover, from Newton's Third Law, the rotational motion of the blades imparts an opposite-in-direction rotational motion to the wind. From the blade section's perspective, the rotational velocity of the wind, ω, is thus perceived as the blade's rotational velocity, Ω, increased by a tangential induction factor of a'_j, such that $a'_j \equiv \omega_j / 2\Omega$. Therefore, the j-th blade section perceives axial and tangential velocity components of $U_\infty(1-a_j)$ and $\Omega(1+a'_j)r_j$, respectively. Notice that these velocity components are in the blade's reference frame, whereas to compute $\vec{u}(\vec{x}_j)$ in Eq. 1.5, we first need to determine these velocity components in the inertial reference frame. In the following subsection, we will introduce the proper geometric considerations to project the velocity components from the blade reference frame onto the inertial reference frame.

The perceived inflow angle at a j-th blade section, ϕ_j, is defined as:

$$\phi_j = \tan^{-1}\left(\frac{U_\infty(1-a_j)}{\left(\Omega + \frac{\omega_j}{2}\right) r}\right) = \tan^{-1}\left(\frac{U_\infty(1-a_j)}{\Omega(1+a'_j)r_j}\right), \tag{1.12}$$

which is the sum of the local angle of attack, α_j, and local blade twist angle, β_j, $\phi_j = \alpha_j + \beta_j$. Finally, the axial and tangential forces acting on each blade section are expected to vary with the free-stream dynamic pressure, $1/2U_\infty^2$, and the rotor's size. Hence, the force coefficients are here defined As:

$$C_{a,j} = \frac{F_{a,j}}{\frac{1}{2}U_\infty^2 R}, \tag{1.13}$$

$$C_{t,j} = \frac{F_{t,j}}{\frac{1}{2}U_\infty^2 R}. \tag{1.14}$$

where R is the rotor's tip radius. Similarly, we expect the circulation Γ_j around each airfoil blade to be a function of the tangential and axial flow velocities. From Buckingham Π-theorem, the circulation is then non-dimensioned by:

$$\Gamma_j^\star = \frac{\Gamma_j}{\frac{U_\infty^2 \pi}{N\Omega}}, \tag{1.15}$$

where N is the number of blades of the rotor.

2
Methodology

2.1. Algorithm and Methods

The implemented LLM approach consists of iteratively solving for the bounded and shed vortex strengths. We start the solution process by establishing the geometry of the rotor and the wake. From the geometric information, we determine the (frozen) induction matrix $A_{i,j}$, such that:

$$A_{i,j} \equiv \frac{\Gamma_i = 1}{u_{\Gamma,i\to j}}. \tag{2.1}$$

Note that A accounts for the influence of the i-th vortex filament on the j-th blade section (or, equivalently, j-th bounded vortex filament). Note also that for a steady-state solution, $\Gamma_i = \Gamma_j$ if $i = j$, i.e., the magnitude of the shed and bounded vortex strengths are equal.

Subsequently to the determination of the static induction matrix A, the iterative solution process ($k = 1, 2, \cdots$) follows as:

1. Calculate the induced velocities: $u^k_{\Gamma,i\to j} = \sum_i A_{i,j}\Gamma^k_i$
2. Determine relative wind speed on control point from Eq. 1.5: $\vec{u}^k(\vec{x}_j) = \vec{u}_{rot}(\vec{x}_j) + \vec{u}_\infty + \sum_i \vec{u}^k_{\Gamma,i\to j}$
3. Determine inflow angle, ϕ^k_j, and angle of attack, α^k_j, from $\vec{u}^k(\vec{x}_j)$
4. Determine the lift force coefficients, $C^k_{l,j}(\alpha^k_j)$, by interpolating α^k_j onto the tabulated polar data; and the bounded circulation, Γ^k_j, from Eq. 1.6: $\Gamma_j = \frac{1}{2}W^{2,k}_j C^k_{l,j}(\alpha^k_j)c_j$, where $W^{2,k}_j = |\vec{u}^k(\vec{x}_j) \cdot \vec{u}^k(\vec{x}_j)|$
5. Calculate change in circulation: $\Delta\Gamma_j \equiv \Gamma^k_j - \Gamma^{k-1}_j$.
6. Update shed circulation values: $\Gamma^k_i = \Gamma^k_i + \Delta\Gamma_j$ if $i = j$.

Steps 1 to 6 above repeat until the solution has converged. In this work, the criterion for convergence in the normalized change in circulation between consecutive time-steps k and $k+1$, $\varepsilon(k)$, is defined here as:

$$\varepsilon(k) \equiv \frac{\Delta\Gamma_{j,\max}}{\max(|\Gamma_j|) + 1} = \frac{\max|\Gamma^k_j - \Gamma^{k-1}_j|}{\max(|\Gamma_j|) + 1}, \tag{2.2}$$

In step 0, i.e., before starting the iterative solution process, the geometry of the blades and wake must be determined. Here, we specify the straight-line vortex filaments by their two nodes (See Fig. 1.2). The first node of the vortex filaments, i.e., the nodes that compose the bounded vortex filaments, are fixed at the respective airfoil's quarter-chord length. The second set of nodes is put $c_j/4$ downstream of the trailing of the respective airfoil, such as illustrated in Fig. 1.2. Notice also that the second set of nodes must follow the airfoil's twist angle, β, to reinforce the Kutta condition:

"*A body with a sharp trailing edge which is moving through a fluid will create about itself a circulation of sufficient strength to hold the rear stagnation point at the trailing edge.*" [3]

In Fig. 1.2, the second set of nodes downstream of the bounded filaments, referred to as "trailing edge nodes", reinforces the Kutta condition since the vorticity filaments are set parallel to the airfoil's chord line. Downstream of the trailing edge filaments (trailing edge nodes), the nodes of the remaining shed vortex filaments are advected by an axial velocity of U_{wake}, such that the wake nodes are allocated onto coordinate $(U_{wake} n_t \Delta t, r\sin(\Omega n_t \Delta t), (r\cos(\Omega n_t \Delta t)))$ downstream of their respective trailing edge node. Therefore, the wake size is set by selecting the number of time steps n_t. In this report, we explore the effects of different advection velocities U_{wake} on the solution, as well as the effects of n_t on the induction factors.

In steps 0 and 1, we must determine the induced velocities using Eq. 1.4. Notice, however, that we introduce singularities in the solution for $r_1, r_2 \to 0$. Hence, we introduce the following "cut-off radius" methodology to avoid numerical instabilities: consider that we replace the singularities $r_1, r_2 \leftarrow 0$ by a small, circular region of radius δ around each vortex filament where the vortex filament strength is constant and equal to that of $r_1, r_2 = \delta$. By doing so, we establish a non-singular solution for cases where x_j is too close to a vortex filament. Figure 1.3 shows the vortex filament-induced velocity as a function of the distance r from the filament before and after the cut-off radius methodology.

In step 1, the induction matrix A, relating the i-th vortex filament with the j-th blade section, is built by setting $\Gamma_i = 1$ and subsequently solving the problem defined in Eq. 1.4. Since A only depends on the geometric characteristics of the rotor and wake, which are frozen-in-time in the current methodology, A can be computed and stored before starting the iterative process. This approach is referred to hear as the "induction matrix approach". Since self-induction is ignored in the current approach, the solution time scales only to the number of vortex filaments.

In step 4, we need to project the velocity vector $\vec{u}_j$ into the axial and tangential directions of the j-th blade section. By definition, the axial and tangential directions at each blade section are relative to the rotor's orientation relative to the free-stream flow and the blade's orientation relative to the rotor. However, since the disk's rotational axis is parallel to the free-stream, $-\infty$, flow, we can determine the axial and tangential flow directions at the blade sections directly from the free-stream flow conditions. For a control point $\vec{x}_j$ at the quarter-chord length of the j-th blade section, the perceived axial, $W_{j,||\vec{u}(-\infty)}$, and tangential, $W_{j,\perp\vec{u}(-\infty)}$, flow velocities are:

$$W_{j,||\vec{u}(-\infty)} = \left(\frac{-1}{|r_j|}\hat{i} \times \vec{x}_j\right) \cdot \vec{u}_j \tag{2.3}$$

$$W_{j,\perp\vec{u}(-\infty)} = \hat{i} \cdot \vec{u}_j \tag{2.4}$$

Finally, in steps 5 and 6, the change in circulation due to the updated values of W_j is used to update the estimation of Γ_j. The iterative process repeats until sufficiently small values of $\Delta\Gamma_j$ are reached.

3
Results

In this report, the rotor's blade sections are modeled as DU95W180 airfoil profiles. The polar data for a DU95W180 airfoil profile are shown in Fig. 3.1 for reference. The pitch angle β of the airfoil sections vary linearly with the distance r from the rotation axis, such that $\beta(r) = -14(1-r) + 2$ degrees, for $r \in [10, 50]$ m, where the rotor tip radius is $R = 50$ m and the rotor root radius is $R_{root} = 10$ m. The chord-length distribution, $c(r)$, also follows a linear distribution law: $c(r) = 3(1-r) + 1$ m. The free-stream flow condition is defined by $U_\infty = 10$ m/s in the rotor's axis direction.

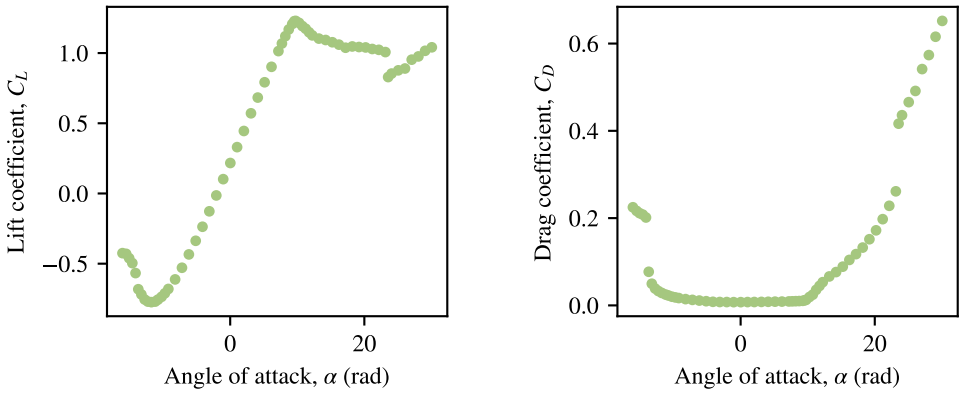


Figure 3.1: Lift and drag coefficients, C_L and C_D, for the DU95W180 airfoil profile. The sections of the rotor's blades modeled in this report consist of DU95W180 airfoil profiles.

3.1. Grid independence analysis

In this subsection, we evaluate the impact of the sizing of LLM's parameters on the model's results. Here, the number of blade sections, N_{BS}, distribution-law for control points, shedding vortex time-step, $\Delta t/T$, and the wake's length, t_f/T, effects on LLM's results are investigated. Notice that we define t_f as the total advection time of the wake, whereas Δt is the time-step between the shedding of two consecutive trailing edge vortices in the wake. The rotor's characteristic time, T, is defined as the rotor's revolution time.

We adopted the thrust and power coefficients, C_A and C_P, as a basis for this grid-independence study. Here, we defined the thrust and power coefficients as:

$$C_A = \frac{\sum_i F_{a,i} \Delta r_i N_B}{\frac{1}{2} U_\infty^2 \pi R^2}, \tag{3.1}$$

$$C_P = \frac{\sum_i F_{t,i} \Delta r_i (r_i \Omega) N_B}{\frac{1}{2} U_\infty^3 \pi R^2}, \tag{3.2}$$

where r_i is the radius of the i-th control point, Δr_i is the i-th blade section and N_B is the number of blades. For the current grid-independence analysis, the rotor is set to rotate at the highest tip speed ratio (TSR) of interest, i.e., TSR$= 8$, where TSR$= \Omega R/U_\infty$.

In Fig. 3.2, we show convergence of the thrust and power coefficients for an increasing number of blade sections, N_{BS}, in the discretization of the rotor's blades. Notice that a cosine-distribution law is applied for seeding the control points over the blade section. The results shown in the figure correspond to the analysis of a single-blade rotor at TSR= 8. For this analysis, the total wake time, t_f, is set to cover ten rotor rotation periods, i.e., $t_f/T = 10$, where $T = 1/(\Omega/2\pi)$. A conservative time-stepping size of $\Delta t/T \approx 0.03$ was selected to advect vortex filaments in the wake. Quick convergence is observed for the LLM concerning the number of control points seeded over the blade. For the current analysis, $N_{BS} = 22$ provides grid-independent results for the selected flow condition at TSR= 8.

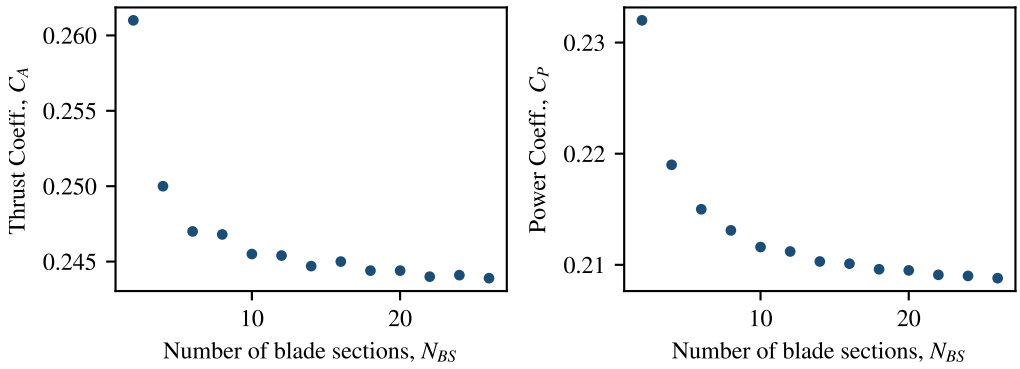


Figure 3.2: Convergence of thrust, C_A, and power, C_P, coefficients for increasing number of control points over a single-blade rotor.

The second parameter study conducted in this report is the distribution law adopted for seeding control points over the blade sections. Two seeding distributions strategies are assessed in this study, namely: i. uniform distribution function and ii. a cosine-distribution function. The uniform distribution function consists of discretizing a blade into elements of equal radial sizing, i.e., $\Delta r_i =$ constant for all the $i = 1, 2, \cdots, N_{BS}$ blade section. The cosine function distribution applies a cosine-projection-based spacing distribution for the blade sections. With the later approach, Δr_i decreases for $r \to R_{root}$ and $r \to R$, and it is maximum for $r = (R_{root} + R)/2$. In Fig. 3.3, we show the circulation distribution, $\Gamma^\star$, and axial induction factor, a, using both uniform and cosine distribution laws for $N_{BS} = 22$. Notice that in both cases, the distributions of $\Gamma^\star$ and a are well captured at the blade's center. However, the higher concentration of control points near the root and tip sections of the blade with the cosine-distribution function allowed for better capturing of the gradients of the evaluated quantitites. Thus, the cosine distribution function will be adopted as the default in this report.

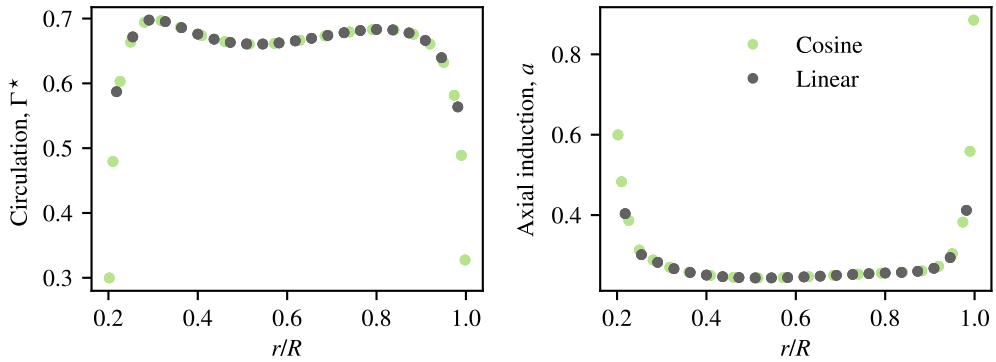


Figure 3.3: Comparison between linear and cosine distribution strategies of control points. Panel to the left shows results for circulation $\Gamma^\star$, and axial induction, a, for the two assessed distribution strategies.

In Fig. 3.4, we analyze the effects of the wake length on the LLM results. We characterize the wake length here by the number of wake rotations extended by the frozen wake, t_f/T, where t_f is the total analysis period. As for the previous analysis, a conservative time-step sizing of $\Delta t/T = 0.03$ is adopted in this analysis. From the results shown in the figure, we conclude that convergence of the

analyses variables is quickly approximated at $t_f/T < 5$, equivalent to a wake extending a distance of only $x/R \approx 2.5$ downstream of the rotor. Any further increases in length of the wake beyond $t_f/T = 10$ ($x/R \approx 5$) results into negligible changes ($< 0.2\%$) of the analysed quantities. Thus, $t_f/T = 10$ is deemed sufficient for wake-length independent results.

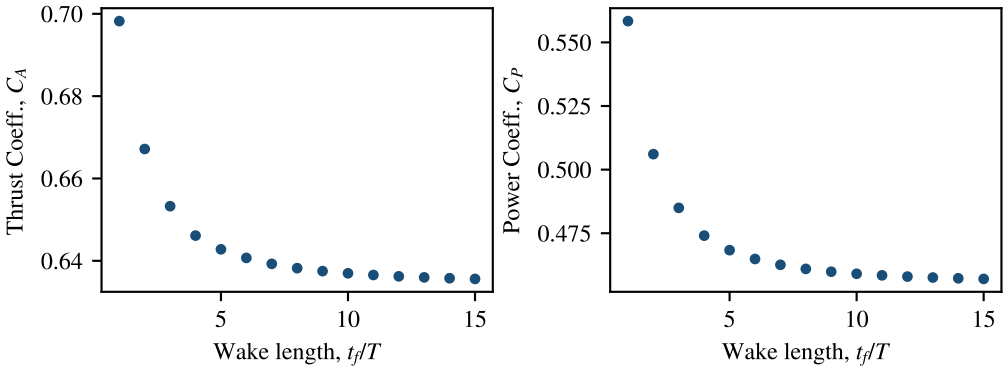


Figure 3.4: Convergence of thrust, C_A, and power, C_P, coefficients for increasing frozen-wake length.

Following the wake length analysis, we study the influence of time-step sizing on the accuracy of the LLM. In Fig. 3.5, we show the circulation and axial induction distributions for different time-step sizings, $\Delta t/T$. Notice that although the analysis carried out in this study is stationary in nature, the adopted frozen-wake methodology requires the seeding of vortex filaments in the wake respecting a shedding-time mechanism, $\Delta x = \Delta t U_{wake}$; where U_{wake} is the wake's advection velocity. In the current context, smaller values of Δt imply higher accuracy in capturing the wake's topology. From the results shown in Fig. 3.5, it is evident that the assessed parameters are independent (circulation, $\Gamma^\star$, and axial induction, a) are insensitive to changes in time-step sizing below $\Delta t/T < 0.1$, i.e., the assessed parameters present differences below 0.2% for $\Delta t/T$ for $\Delta t/T < 0.1$. Hereafter, simulations are run with $\Delta t/T = 0.1$.

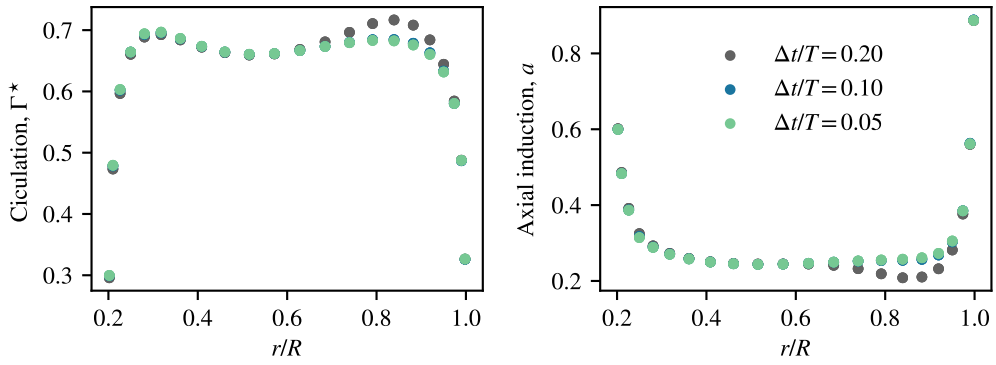


Figure 3.5: CDistribuition of circulation $\Gamma^\star$, and axial induction, a, for different frozen-wake time-step sizings $\Delta t/T$.

In Fig. 3.5, we showed the effects of the time-step sizing on the distribution of $\Gamma^\star$ and a. Subsequentially, we show in Fig. 3.6 the distribution of the same variables for different wake advection speeds, $U_{wake} = U\infty(1 - a_w)$. More precisely, we investigate here the effects of the wake induction factor, a_w, on the LLM's results. Notice also that in the frozen-wake methodology, the x-wide spacing of the vortex filaments is simply given by $\Delta x = \Delta t U_{wake}$. Thus, large values of U_{wake} imply high convective velocities, hence a more "stretched out" wake topology. On the other hand, low values of U_{wake} imply a more "compressed" wake geometry.

Here, we set a constant speed for the advection of the wake that is constant and inversely proportional to a wake induction parameter of a_w. We explore the following values for a_w: 0, 1/3, 2/3. For $a_w = 0$, the wake is advected at the free wind's speed. For $a_w = 1/3$, the wake is advected at the convection speed corresponding to Betz's limit for maximum possible power production for a disk actuator surface.

For $a_w = 2/3$, the wake is convected at the infinitely-downstream speed of a stream tube at Betz's limit. We show the results for this analysis in Fig. 3.6, where we also included results from a Blade Element Momentum (BEM) theory code with Prandtl's tip and roor-losses corrections. The same free-stream flow conditions are adopted in both LLM and BEM codes.

From the results shown in Fig. 3.6, it is evident that $a_w = 1/3$ yields the best approximation of the Lifting-line model to a BEM model. For $a_w = 0$, the wake becomes "stretched-out" in the streamwise direction, and the resulting induction at the rotor's blade decreases as the vortex filaments move further downstream from the bounded vortex filaments. For $a_w = 2/3$, the opposite effect is observed. Since the wake becomes more compact, the average distance between the vortex filaments and the bounded vortex decreases, thus decreasing the induction factor. The blade perceives the higher wake-induction factor as an overall higher angle of attack, thus the higher circulation values for increasing a_w.

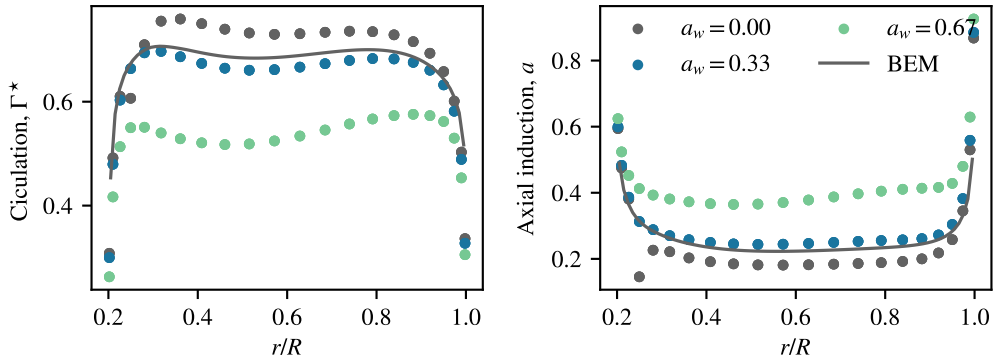


Figure 3.6: CDistribuition of circulation $\Gamma^\star$, and axial induction, a, for different frozen-wake induction factors a_w.

3.2. Spanwise distribution of the parameters

In the previous subsection, we evaluated the impact of the discretization strategies on the solution results of the current LLM method. In this subsection, we adopted the best (i.e., higher-accuracy) strategies and discretization sizings concluded from the previous subsection to evaluate the performance of the rotor benchmark at the default flow conditions. Here, the default flow conditions are given by $U_\infty = 10$ m/s, TSR= 8. The wake is discretized in 22 elements in the spanwise direction using a cosine distribution function for the seeding of control points. The advection time is $\Delta t/T = 0.1$ for 400 time-steps, corresponding to $t_f/T = 10$ periods of rotation of the rotor. Here, the rotor is composed of three blades designed with a twist angle distribution of $\beta(r) = -14(1-r)+2$ degrees and chord-sizing a distribution of $c(r) = 3(1-r)+1$ m. The wake induction factor, a_w, is set to $a_w = 1/3$. The resulting wake-rotor problem geometry is shown in Fig. 3.7.

We show in Fig. 3.8 the spanwise distributions of the angle of attack, α, inflow angle, ϕ, circulation, $\Gamma^\star$, and force coefficients for the default flow case, i.e., $U_\infty = 10$ m/s and TSR= 8. We also overlay results obtained with a BEM code for the same flow conditions for reference. The circulation distribution shows that the most significant contribution to lifting force production occurs between $r/R = 0.3$ and 0.9. At the tip and root ends of the blades, pressure from the suction and pressure sides equate, and no lift is produced. The no-lift condition at the two ends of the blades translates into zero circulation in the first and last control points in the spanwise direction. The equilibrium solution, as shown in the bottom-left panel of the figure, is a quick drop in circulation towards R_{root} and R.

Some interesting remarks about the dynamics of horizontal axis wind turbines can be drawn from the force coefficient distributions. For instance, despite the smaller blade sections composing the extrados of the blades, significant thrust (axial force) is induced at this region. This fact is a simple consequence of the high perceived wind speeds in the outer rings of the stream tube. Moreover, the tangential force coefficient (for the given blade design) is approximately constant along the blade for the force coefficients normalized by the blade radius, R. Moreover, the axial forces acting on the blade are considerably larger in magnitude than the obtained tangential force components.

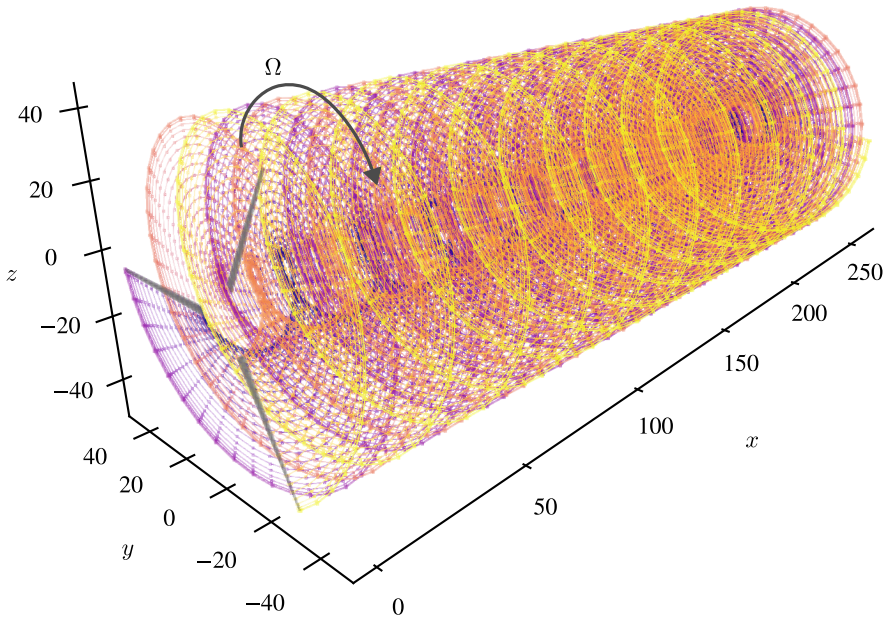


Figure 3.7: Blades and wake geometries of the 50-meters-radius rotor benchmark.

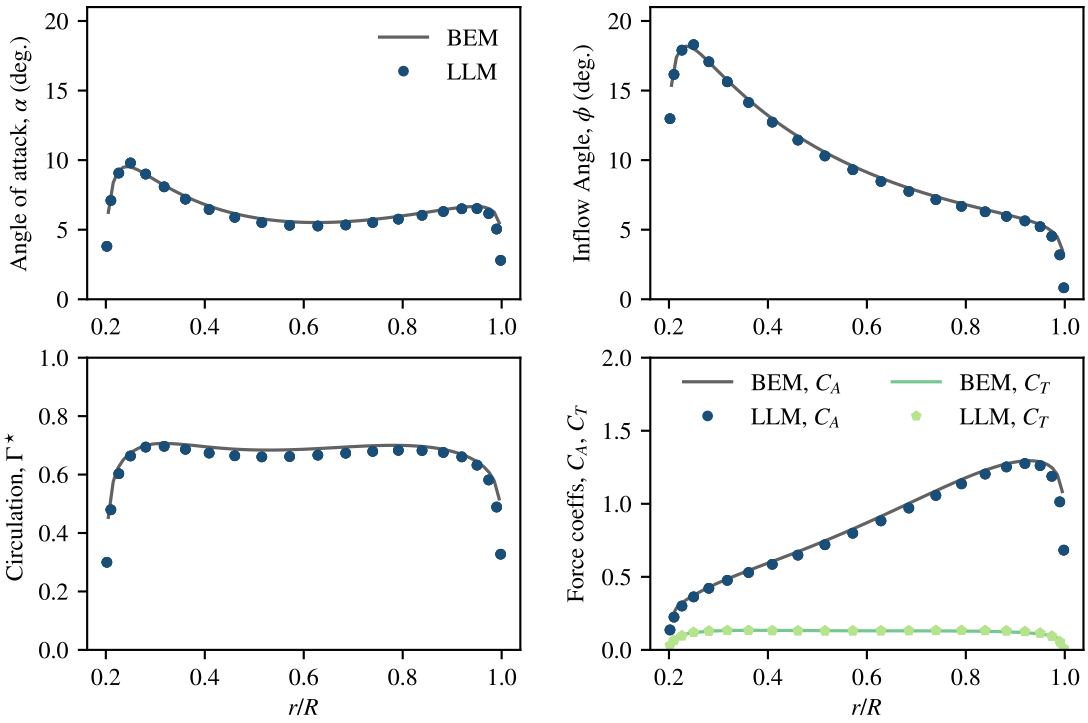


Figure 3.8: Spanwise distributions of angle of attack, inflow angle, circulation and force coefficients for the 50-meters-radius rotor at TSR= 8. Results from a BEM code with the same flow conditions are overlaid for reference. Here, C_A is the axial (or thrust) coefficient and C_T is the tangential force coefficient.

Lastly, we investigate the effects of the tip-speed ratio, TSR, on the performance of the benchmark wind turbine. We recall here the expressions for thrust and power coefficient in Eqs. 3.1 and 3.2, respectively. Notice that for this analysis, the wake induction a_w is kept constant and equal to 1/3. Thus the accuracy of this analysis is limited.

Nonetheless, the expected increase in thrust and power coefficients with TSR characteristic of disk

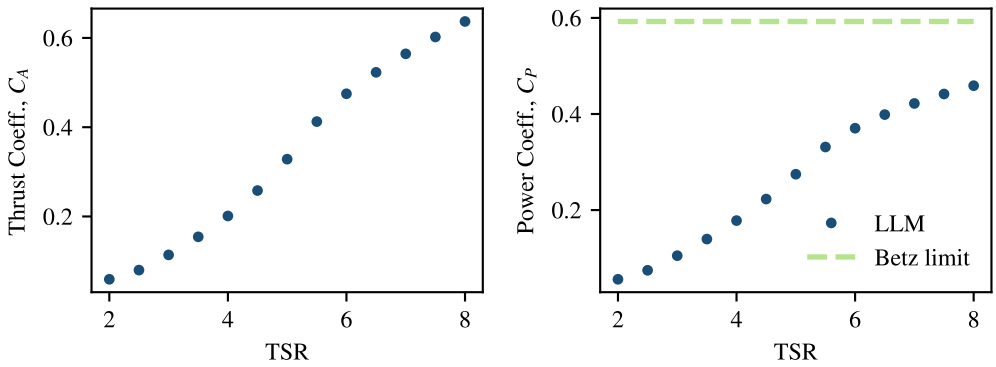


Figure 3.9: Net thurst, C_A, and power, C_P, coefficients at the totor's shaft for different TSR values. Betz limit for ideal (non-rotating wake) rotor model is overalid for reference.

rotor models is observed. Notice that at TSR= 8, the power coefficient gets closest to the Betz limit of 16/27. However, beyond this point of maximum power output, the rotor enters a "heavily-loaded" state, where the induction factors become too high due to the high blade solidity (see [4]), and the power output of the rotor is expected to decrease.

4
Conclusions

This report presents a Lifting-line theory code and benchmark results obtained with the same code for a 50-meter-radius wind turbine rotor subjected to different wind flow conditions. We adopted a frozen-wake approach to constructing the blades and wake geometry in the presented Lifting-line code. In the frozen-wake methodology, the distribution of vortex filaments in the wake is pre-established as a user input, such that the streamwise spacing between consecutive shed vortex filaments is a function of the user-defined wake convection speed. This report evaluated the effects of i. total wake length, ii. wake streamwise and spanwise discretization sizings, iii. wake convection speed, and iv. distribution functions for the control point over the rotor's blades. We concluded from this analysis that lifting-line models are more accurate for longer total-wake lengths. However, wake-induced velocities are sufficiently well captured for wake lengths covering as little as ten rotor revolutions. Moreover, the solution results are also satisfactorily grid independent for blades discretized into approximately twenty blade sections. It was also observed that an overall induction factor of 1/3 for the advection of the wake filaments allows for the best results in the implemented frozen-wake lifting-line model.

In terms of wind-turbine aerodynamics, it is concluded from the current benchmark that power production increases in the blade's spanwise direction due to the higher perceived wind velocities for increasing spanwise coordinates. It was also observed that maximum axial induction happens at the blade's tip and root sections due to tip and root losses. Power production was also more maximum for tip-speed ratios of eight. For tip speed ratios below eight, low angles-of-attack yield sub-optimum aerodynamic loadings. Higher tip speed ratio values than eight were not assessed in this report.

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